

Pooled OLS: $y_{it} = \alpha + \beta \cdot x_{it} + u_i + e_{it}$

$\uparrow$ one intercept
 $\uparrow$ one slope
 $\uparrow$ household specific preferences, etc.
 $\uparrow$ idiosyncratic error

$$v_{it} = u_i + e_{it}$$

$$\text{Cor}(v_{it}, v_{is}) = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_e^2}$$

Fixed Effects
FE:

$$y_{it} = \alpha_i + \beta x_{it} + v_{it}$$

$\uparrow$
one intercept per unit.

FE Within estimator

unabsorbing
as is

$$(x_{it} - \bar{x}) = (x_{it} - \bar{x}_i) + (\bar{x}_i - \bar{x})$$

Within

Between

3.78%

$$T_{xx} = W_{xx} + B_{xx}$$

$$b_{\text{Pooled}} = \frac{w_{xy} + B_{xy}}{w_{xx} + B_{xx}} = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

$$b_{FE} = \frac{w_{xy}}{w_{xx}} \quad (\text{Ken within variance used})$$

LM : Is there a "preference" effect

$$H_0: \sigma_u^2 = 0 \quad (\text{Pooled OK}) \quad H_1: \sigma_u^2 > 0$$

$$LM \sim N(0, 1) \quad CV_{5\%} = 1.645$$

$$b_{RE} = \frac{w_{xy} + w B_{xy}}{w_{xx} + w B_{xx}}$$

$$w = \frac{\sigma_e^2}{\sigma_e^2 + T \cdot \sigma_u^2}$$

$$\lambda = 1 - \sqrt{w}$$

$$\sigma_u^2 = 0 \quad \lambda = 0 \quad \text{Pooled OLS}$$

$$T\sigma_u^2 \gg \sigma_e^2 \rightarrow \lambda \approx 1 \quad FE$$

Hausman test: FE or RE?
(endogeneity)

$H_0: \mu_i$ is unrelated to x .

$\text{cov}(\mu_i, x) = 0$ | EXOGENOUS
↑
"RANDOM" EFFECTS

	FE	RE
H_0 TRUE	CONSISTENT	CONSISTENT AND EFFICIENT
H_0 FALSE	CONSISTENT	INCONSISTENT

CATCH 22

what if the ^{true} Data Generating Proc.

$$y_{it} = \alpha_i + \beta \cdot x_{it} + \lambda \cdot z + v_{it}$$

↑
time invariant

Hausman only test β between
FE & RE.